Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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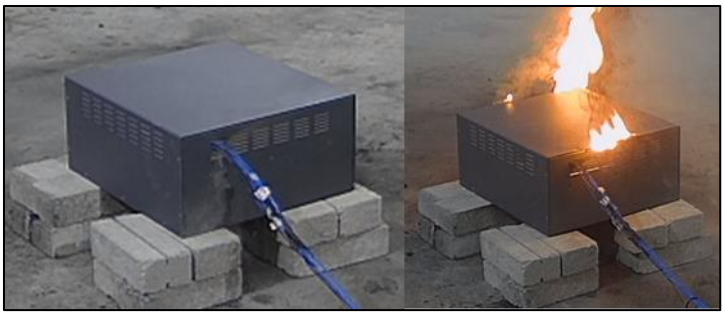
2. Technology Overview

Title of Proposed Technology	**Integrated safety design combining potting-based thermal isolation and flame suppression to mitigate battery pack thermal runaway propagation**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		*Electrical · Electronics*	
	Medium Category	*Small Category*	*Medium Category*	*Small Category*
	Heavy Electrical Equipment	*Energy Storage Devices*	*Batteries*	*Secondary Batteries*
Concept of Proposed Technology	**Core Concept** This technology is an integrated thermal runaway response technology that (1) suppresses heat diffusion and mechanical damage with full-module potting and thermal isolation design to ensure battery pack safety, and (2) controls vent gas emissions with an inductive exhaust path and flame suppression structure to reduce the risk of flame. The goal is to minimize chain reactions and secondary damage in the event of thermal runaway, while simultaneously ensuring mass production and durability.  **Figure. Current results of our thermal runaway test**			

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	Key Functions 1) Inhibiting heat diffusion: reducing air gaps through potting and controlling heat transfer paths with heat-insulating/isolating structures. 2) Mitigating the Effects of Thermal Runaway: Delaying/Suppressing Chain Reactions with Isolation and Confinement Designs. 3) Gas Emission Control: Reduce vent gas leakage and stabilize the exhaust direction with the plenum/duct concept. 4) Flame Suppression: Flame-blocking/quenching structures reduce the risk of flame propagation and flame eruption. 5) Durability and environmental stability: Potting combination improves vibration/shock resistance and moisture/dust stability. **Potential Applications** 1) AMR/AGV and logistics robots 2) Service and Industrial Robots 3) Small electric mobility 4) Humanoid Robot 5) ESS/UPS backup power
Rationale for Joint R&D Project	**Critical Technical Barriers** This R&D project requires demonstration of battery pack–level thermal runaway mitigation performance, which demands integrated optimization of materials (potting and thermal insulation), structural safety design (induced emissions and flame suppression), processes, and testing/verification. As the technology targets export and global market entry, it is essential to secure an internationally reliable verification framework based on reproducible data, test protocols, and global evaluation standards, which necessitates joint research with leading international institutions. **Why Global Collaboration?** 1) It is necessary to integrate the analysis and verification capabilities of leading overseas research institutions with the design and mass production capabilities of domestic SMEs to translate research outcomes into mass-producible products. 2) It is necessary to optimize the potting-based thermal isolation and flame suppression/induction exhaust structure, including the reproducible process window (mixing, injection, curing, and void management). 3) It is necessary to establish testing methods and acceptance criteria (safety, durability, and environment) that conform to international practices to facilitate future certification preparation and product comparison and verification. 4) It is necessary to jointly develop outputs that can be immediately utilized for commercialization and export, such as UL certification, integrated prototype guides, QA/QC checkpoints, and mass production transition criteria.
Key Words	1. Thermal Runaway Mitigation 2. Full-Module Potting 3. Controlled Venting (Plenum/Duct) 4. Flame Suppression / Flame Arrestor 5. Flame Arrestor (Quenching)

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

Goal of this research and development project is to develop and validate a battery pack safety design technology that integrates potting-based thermal isolation technology, inductive exhaust paths, and flame suppression (quenching) structures to simultaneously suppress heat spread and the risk of flame in the event of thermal runaway in a battery pack. To achieve this, the following technologies will be developed

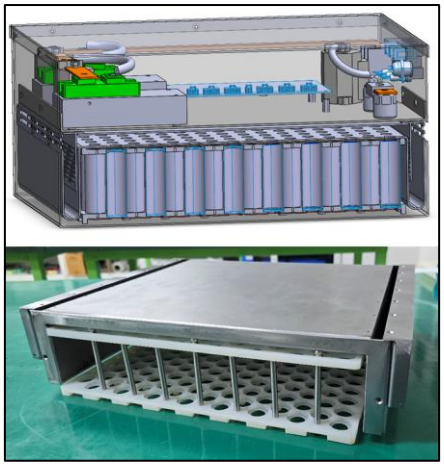

Figure. Current progress results

Detailed Objectives

Core concepts of the technology to be developed: The technology to be developed in this project integrates the following two axes.

1) Potting-based thermal isolation design technology.
2) Apply potting materials to minimize air gaps and unintentional heat transfer paths around cells and incorporate heat-insulating/heat-blocking elements in necessary sections to delay heat transfer to adjacent cells.
   - Purpose: To change the “structure in which heat spreads quickly” to a “structure in which heat spreads poorly”, and especially to design and control the temperature gradient and heat transfer path around the cell where thermal runaway occurs. At the same time, the application of potting has the additional effect of reducing the relative displacement of cells/components from a mechanical point of view, thereby enhancing vibration and shock resistance, and from an environmental point of view, reducing the risk of moisture/dust ingress.
3) Vent-gas control and flame suppression design technology.
   - The high temperature and high-pressure vent gas generated during thermal runaway is not randomly leaked, but is guided to the **intended exhaust path (plenum/duct concept)** to stabilize the discharge direction.
   - Since flames may accompany or propagate during the discharge process, a

flame suppression (quenching) structure is applied to the discharge section to reduce the risk of flame eruption and external flame propagation.

- This changes the path from "thermal runaway → gas/flame spreads internally/laterally → chain damage" to "occurrence → controlled discharge to the flow → reduced risk of fire".

Specific development technology items (detailed goals): In this research and development, the specific technologies below are developed in the basic stages of design, production, and verification.

1) Potting/thermal insulation materials and application design technology.
   - Select a group of potting material candidates suitable for the purpose of thermal runaway mitigation and compare application characteristics such as thermal conductivity/heat resistance/flame retardancy/adhesion/viscosity/curing behavior to reflect them in the design.
   - Establish a design that minimizes voids and ensures process reproducibility, including potting application area (full/partial), thickness, filling direction, and degassing (air removal) strategy.
   - Optimize the thermal barrier strategy through a combination of potting and thermal insulation elements (thermal insulation sheets/bulkheads, etc.).
2) Induction exhaust path (plenum/duct) and sealing (leakage prevention) design technology.
   - Design the flow path shape, joint structure, and sealing interface so that the thermal runaway gas is guided to the exhaust port without leaking laterally from the inside.
   - During the design process, a structural plan that satisfies both function and mass production (DFM) is derived by considering pressure rise, flow rate, flow resistance, interference between internal components, and assemblability.
   - Define design margins and inspection items to prevent clogging of the exhaust path, unintentional leakage, and functional degradation due to assembly deviations.
3) Flame suppression (quenching) structure application technology.
   - Design the application method of flame suppression elements such as metal mesh/porous media and optimize the balance between flame suppression performance and pressure loss when combined with the exhaust path.
   - Define component specifications and fastening methods by reflecting considerations for heat resistance, structural stability, and long-term durability (corrosion/dust/contamination) in high-temperature gas environments.
4) Establishing a process window and establishing mass production quality standards (QA/QC).
   - Set standard conditions for the potting process (mixing-injection-curing-inspection) and establish a process window that manages key variables such as temperature/time/viscosity/defoaming conditions.
   - By defining inspection items (e.g. voids, filling rate, leakage, discharge assembly status) and acceptance criteria for mass production application, we

	aim to create a quality system that can be reproducibly implemented on the manufacturing line. 5) Establishment of testing and verification protocols and evaluation criteria (based on international trust). ● Systematize test items and judgment criteria from the perspectives of safety (heat/flame/gas), durability (vibration/shock), and environment (moisture/dust), including pack tests based on thermal runaway single-cell triggers. ● Manage variability in results through repeated testing and standardized data. ● Collection/recording systems (sensor placement, log standards, video recording, etc.) to enable performance comparisons when design changes occur. ● Finally, a verification package is built in the form of "design rules + process conditions + test methods + acceptance criteria" to pass UL certification. Expected Outputs (Deliverables) : At the end of this project, the following tangible deliverables will be provided. ● Integrated safety design architecture (porting/thermal isolation + inductive exhaust + flame suppression) design and prototype. ● Potting/assembly process standard conditions and process window documentation. ● QA/QC checkpoints and inspection items/criteria (for mass production). ● Test and verification protocols (safety, durability, and environment) and reproducible data (to ensure international reliability). ● Integrated guide for mass production (design, process, and verification package).
R&D Target Specifications	In the evaluation of a pack system based on a single-cell initiation thermal runaway, The goal is to simultaneously satisfy thermal diffusion suppression, flame hazard reduction, gas emission control and mass production reproducibility. 1) Thermal runways occur when Adjacent cells reach 200°C. Time ≥ 10 minutes. ● After a single cell trigger, adjacent cell surface/can temperature (TC) is measured. 2) Suppression of fire spread to other components until test termination ● As determined based on temperature and gas events 3) Sustained Flame at External Vent Outlet (>1 s): 0 occurrences ● High-speed camera and visual inspection, supported by event logs 4) Vent Flame (if any): Length ≤ 100 mm, Duration ≤ 1 s ● Video-based measurement of flame length and duration 5) Directed Discharge Ratio ≥ 80% (Side/Unintended Leakage ≤ 5%) ● Based on particulate collection and flow rate comparison, or gas tracer techniques

(max 10 pages)

	6) External Surface Temperature at 50 mm ≤ 150 °C ● Measurement using temperature sensors mounted on the housing outer wall 7) Potting Void Ratio ≤ 1.0% ● One of the following methods may be selected: cross-sectional analysis, CT scanning, or gravimetric method 8) The pass rate for all key safety metrics listed above shall be ≥ 90%, based on testing of 10 prototype units ● Repeated testing under identical conditions 9) Post-Vibration/Shock Vent Path Integrity: No Leakage/Blockage, No Safety Performance Degradation ● Functional verification after durability testing, followed by re-confirmation of key safety metrics
R&D Plan	This project is based on the premise that achieving a successful battery pack thermal runaway test in a single attempt is difficult. Therefore, rather than aiming for a “one-time pass,” the project will be carried out through an iterative process of repeated testing, root cause analysis, and implementation of improvements to progressively enhance performance. The testing process will be conducted in three stages: bench level, sub-module level, and pack system level. As pack-level system testing involves high costs, the number of such tests will be increased incrementally on a year-by-year basis. In the early phase, the primary objective is learning and elimination of failure modes. In the later phase, the focus shifts to achieving a high pass rate and ensuring reproducibility. 1) Research Approach ● Application of the DBTL methodology: Repeated cycles of Design → Build → Test → Learn ● Three-stage testing framework: Bench-level testing, Sub-module testing, Pack system testing ● Change management (ECO: Engineering Change Order) 2) Key Work Packages (WPs) and Methodology ● WP1. Definition of Requirements and Establishment of the Test Framework ● WP2. Optimization of Potting-Based Thermal Isolation ● WP3. Development of Thermally Induced Venting Paths and Sealing ● WP4. Development of Flame Suppression (Quenching) Structures ● WP5. Iterative Development of Integrated Prototypes (Revisions) and System-Level Testing ● WP6. Packaging for Mass Production and Export 3) Year-by-Year Test Plan (Reflecting Practical Constraints) Year 1 (Months 0–12): Functional Feasibility ● Bench-level testing: 1–2 tests per month ● Sub-module testing: 2 tests per year

	● Pack system thermal runaway testing: 1–2 tests per year Year 2 (Months 13–24): Target Performance Convergence and Initial Reproducibility ● Bench-level testing: ≥ 2 tests per month ● Sub-module testing: 1 test per quarter (4 per year) ● Pack system thermal runaway testing: 1 test per quarter (4 per year) Year 3 (Months 25–36): Validation and Establishment of Export-Grade Confidence ● Bench-level testing: 1–2 tests per month ● Sub-module testing: 1 test per quarter (4 per year) ● Pack system thermal runaway testing: 9–12 tests per year 4) Final Deliverables (Summary) ● An integrated safety design prototype incorporating potting-based thermal isolation, guided venting, and flame suppression ● Defined process windows and QA/QC checkpoints ● Test protocols, pass/fail criteria, and repeatability datasets serving as evidence of export-grade reliability

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1) Technical Contributions ● Proposal of design directions based on analysis of thermal, flow, and flame behaviors during thermal runaway events. ● Quantification of the impact of key design variables, including potting/thermal barriers, venting and sealing, and quenching structures. ● Root cause analysis of major failure modes (leakage, blockage, pressure, and flame propagation) and recommendation of improvement priorities. 2) Scope of Research ● Development of test scenarios, instrumentation plans, and a validated verification roadmap. ● Support for model- and data-driven design optimization, including trade-off analyses. ● Support for the establishment of internationally credible test protocols and pass/fail criteria. 3) Support Provided ● Deployment of battery safety and thermal management experts (PI and researchers). ● Provision of analysis methodologies and data processing templates (decision logic and report formats). ● Delivery of practical design guidelines for development, fabrication, and testing. 4) Specific Tasks ● Support for defining KPI- and UL-based thermal runaway pass criteria (thermal propagation suppression, flame suppression, guided venting, and pressure safety). ● Analysis of bench-, sub-module-, and system-level test data and derivation of improvement actions. ● Support for sourcing potting compounds and other critical materials not readily available in Korea. ● Support for assembling validation packages for international submission (repeatability datasets and report templates).
Expected R&R of Korean SME Partner	1) Development Activities ● Integrated design and prototype fabrication with iterative revisions (Rev. A → B → C → D). ● Implementation of potting processes (mixing, degassing, dispensing, curing) and thermal barrier configurations. ● Fabrication and assembly of guided venting paths and sealing structures with ensured assemblability and mitigation of leakage and blockage risks. ● Modularization of flame suppression structures, including layered configurations, fixation methods, and serviceability/replacement considerations. 2) Testing and Verification Activities ● Repeated bench-level testing (flow rate, pressure, leakage, quenching performance, and void ratio). ● Sub-module testing for integrated functionality verification and risk elimination. ● Preparation and execution of pack system thermal runaway tests (or coordination with

	outsourced test facilities), including collection of evidentiary materials (video recordings and data logs). ● Durability and environmental testing (vibration, shock, temperature, and humidity) with pre- and post-test performance comparison. 3) Resource Allocation ● Deployment of design, manufacturing, and quality personnel (mechanical design, process engineering, and QA). ● Prototype fabrication resources, including dispensing, degassing, and curing processes, as well as assembly jigs. ● Budget and schedule management for the use of external test laboratories. 4) Collaboration Activities ● Operation of regular technical review meetings to share test results and finalize subsequent design and test plans. ● Operation of a structured change management (ECO) system linking design changes, test results, and pass/fail decisions ● Establishment of QA/QC checkpoints and mass production transition criteria, including inspection items and allowable tolerances

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** 1) Thermal Runaway Propagation Mitigation ● Delayed heat transfer to adjacent cells through potting-based thermal isolation and thermal barrier structures. ● Reduced risk of chain reactions and minimized damage extent by promoting localized termination at the cell or module level. 2) Guided Venting and Flame Risk Reduction (Venting & Flame Safety) ● Reduced vent gas leakage and stabilized discharge direction through guided venting paths. ● Mitigation of sustained flames and flame jet risks through flame suppression (quenching) structures. 3) Establishment of a Verification Framework (Export-Ready Evidence) ● Securing validated test protocols, pass/fail criteria, and repeatability datasets. ● Construction of comprehensive verification package linking design rules, process windows, QA/QC checkpoints, and test results. **Commercialization Potential** 1) Readiness for Mass Production ● Establishment of a robust potting process window, including mixing, dispensing, curing, and void control. ● Definition of QA/QC checkpoints and inspection items (leakage, voids, and assembly condition). 2) Scalability and Applicability Across Markets ● Applicability to multiple markets, including ESS/UPS systems, data center backup power, robotics (humanoid robots, AMRs, and AGVs), and small mobility platforms. ● Feasibility of derivative designs to accommodate variations in pack configuration and size. 3) Enhanced Export Competitiveness ● Achievement of UL certification through successful UL thermal runaway testing. ● Improved global credibility by securing verification evidence that meets the expectations of overseas customers and certification bodies. **Industrial & Economic Impacts** 1) Minimization of Fire Damage in Large-Scale Systems such as ESS ● Targeting safe behavior in which accident outcomes are limited to localized damage at the cell or module level, rather than complete destruction of entire facilities or racks. ● Reduction of risks related to human safety, asset loss, and operational downtime. 2) Reduction of Total Cost of Ownership (TCO) ● Expected decreases in downtime, maintenance expenses, safety response costs, and quality-related losses. 3) Strengthening Global Market Entry for Domestic SMEs ● Enhanced productization and export readiness of small and medium-sized enterprises through integrated design, process, and verification package.

6. References

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